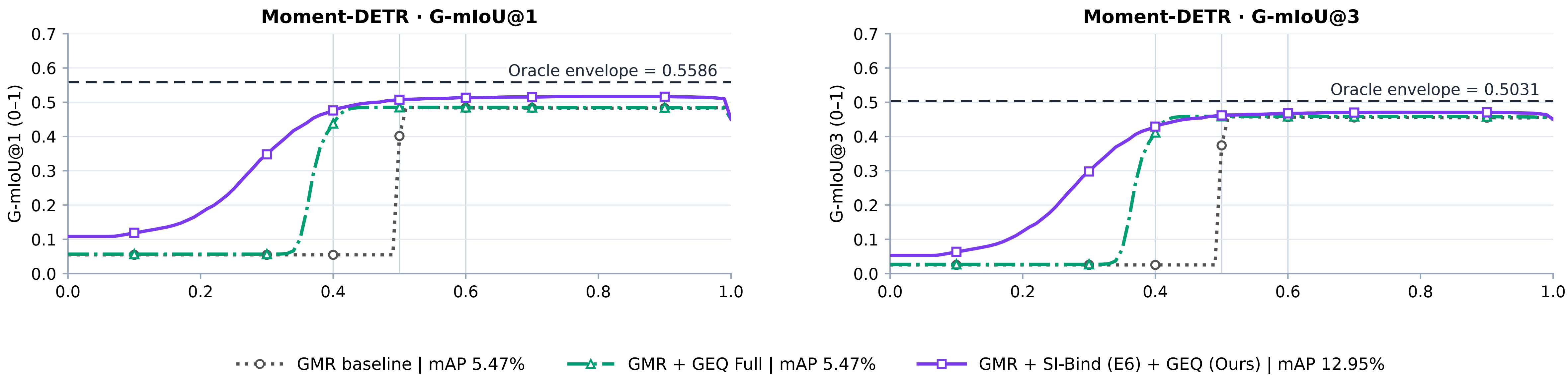
# SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E6) + GEQ

Standard-trained, fixed Val-selected best checkpoints → SoccerGMR Full-Decodable Test (2,970 queries)

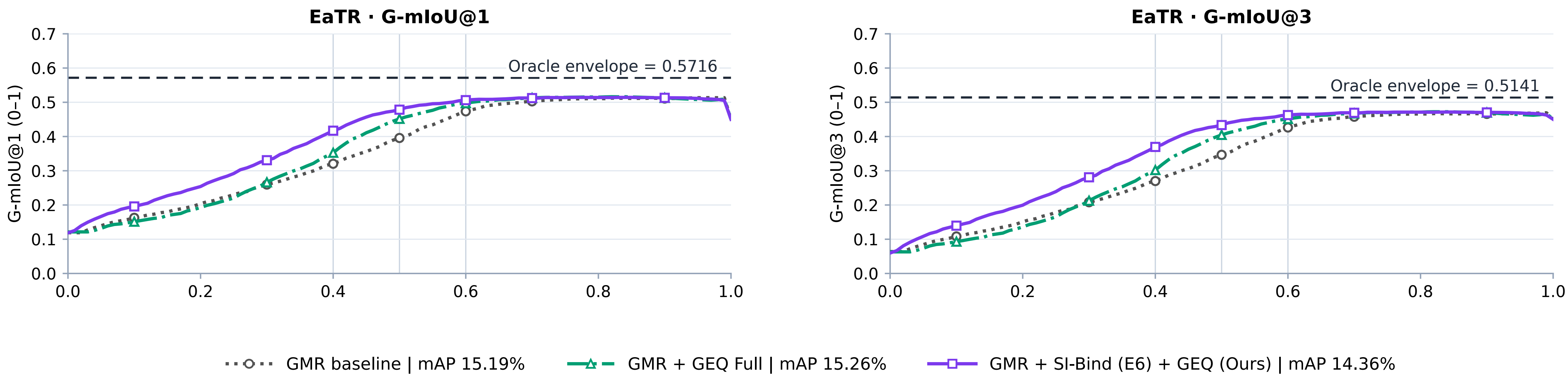
Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]



**ΔmAP vs GMR: +7.48 pp (vs GEQ: +7.48 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +18.56 pp

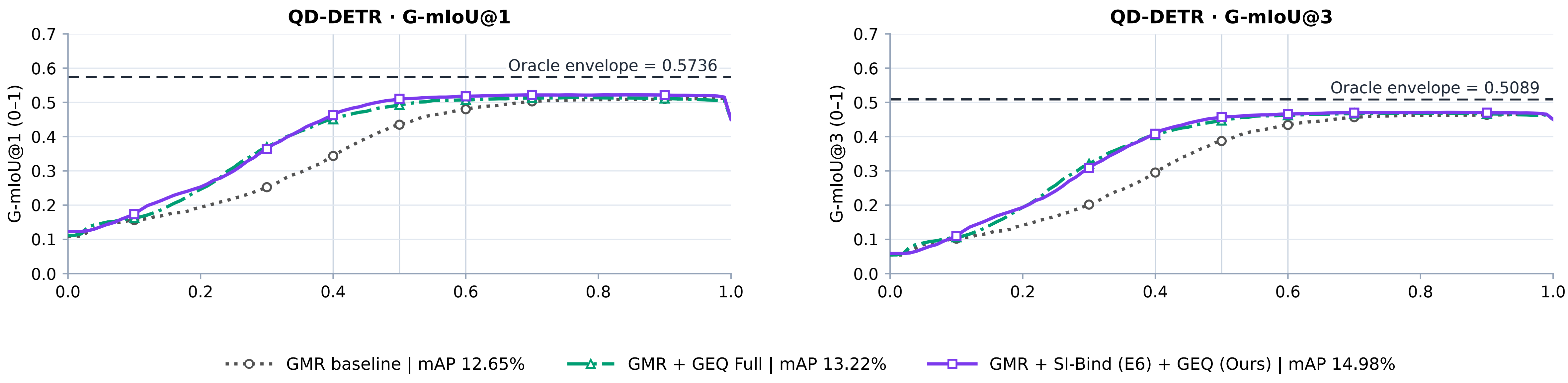
ΔG3 mean vs GMR (.4/.5/.6): +16.74 pp



**ΔmAP vs GMR: -0.83 pp (vs GEQ: -0.90 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +7.07 pp

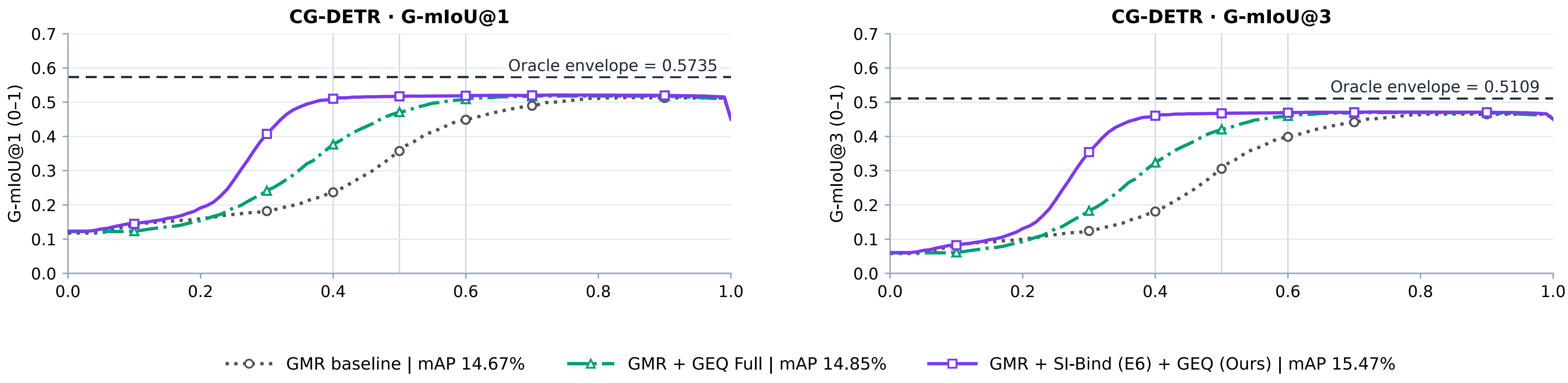
ΔG3 mean vs GMR (.4/.5/.6): +7.43 pp



**ΔmAP vs GMR: +2.33 pp (vs GEQ: +1.76 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +7.75 pp

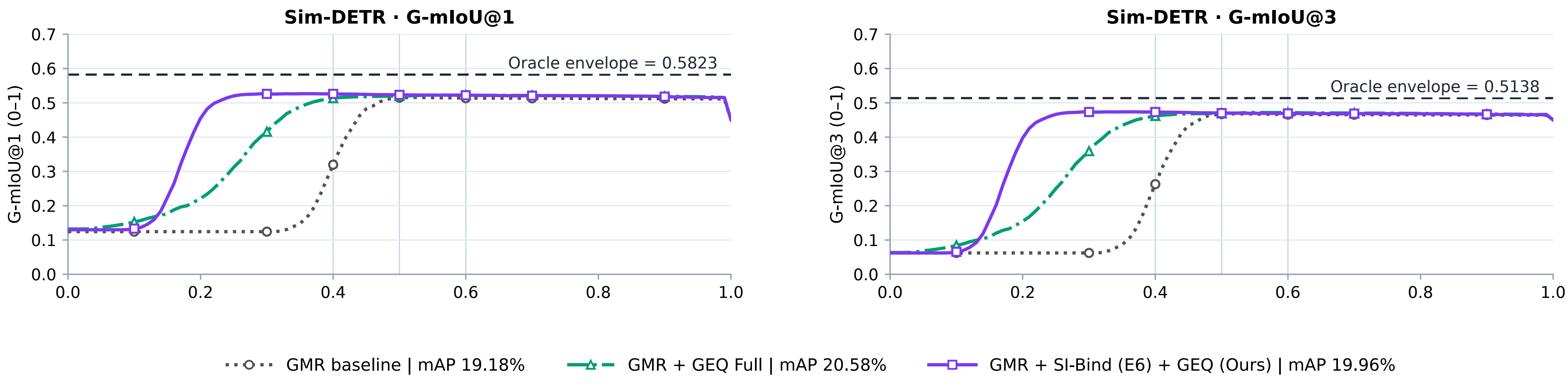
ΔG3 mean vs GMR (.4/.5/.6): +7.17 pp



**ΔmAP vs GMR: +0.80 pp (vs GEQ: +0.62 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +16.77 pp

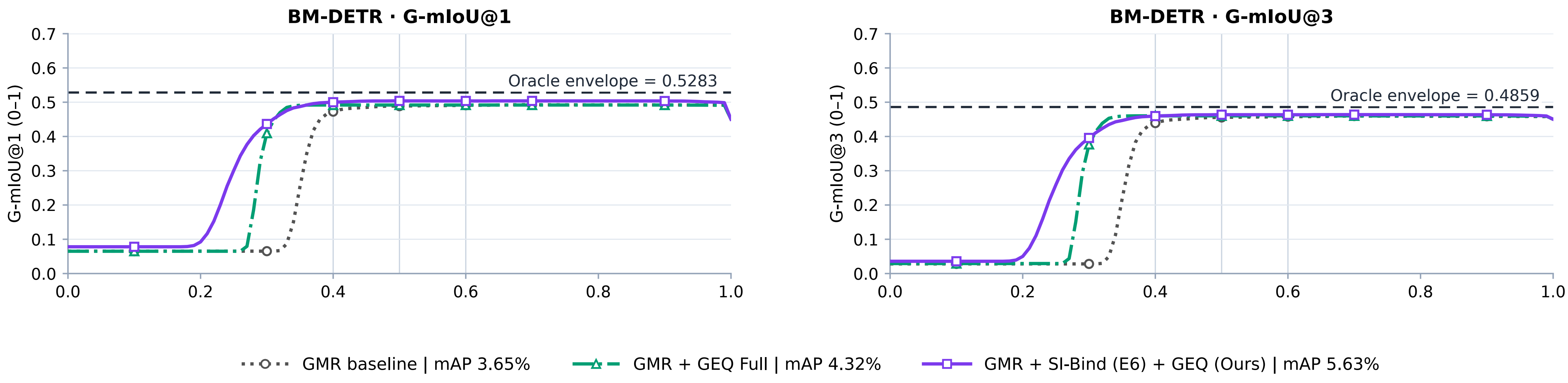
ΔG3 mean vs GMR (.4/.5/.6): +17.07 pp



**ΔmAP vs GMR: +0.78 pp (vs GEQ: -0.62 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +7.44 pp

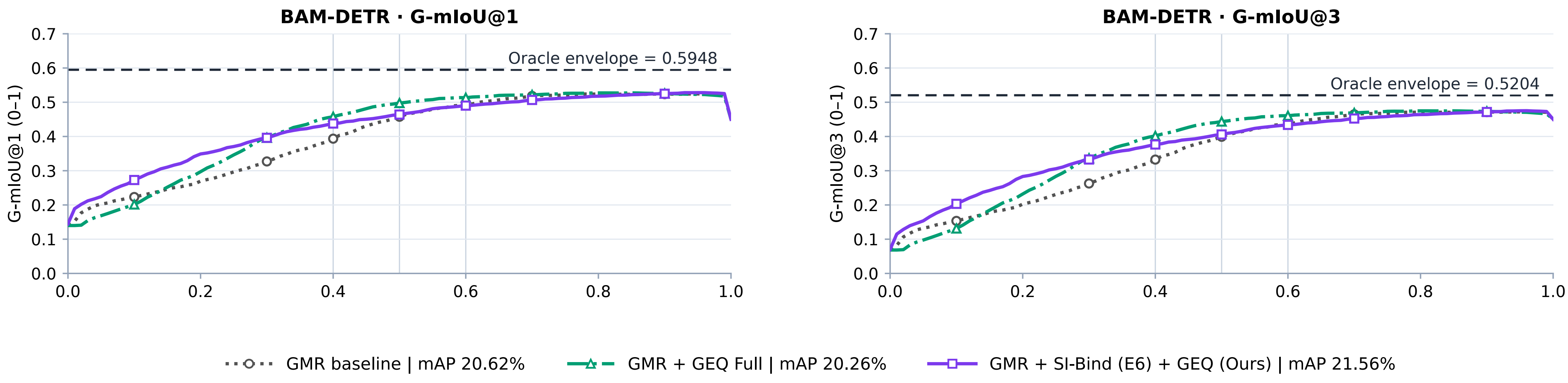
ΔG3 mean vs GMR (.4/.5/.6): +7.23 pp



**ΔmAP vs GMR: +1.98 pp (vs GEQ: +1.31 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +1.85 pp

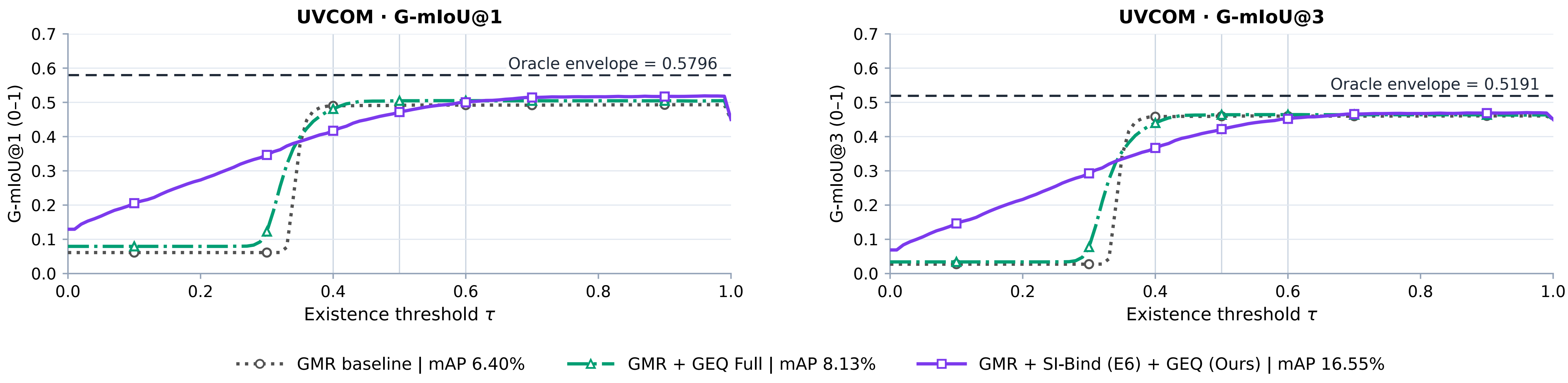
ΔG3 mean vs GMR (.4/.5/.6): +1.13 pp



**ΔmAP vs GMR: +0.94 pp (vs GEQ: +1.30 pp)**

ΔG1 mean vs GMR (.4/.5/.6): +1.59 pp

ΔG3 mean vs GMR (.4/.5/.6): +1.58 pp



**ΔmAP vs GMR: +10.15 pp (vs GEQ: +8.42 pp)**

ΔG1 mean vs GMR (.4/.5/.6): -2.81 pp

ΔG3 mean vs GMR (.4/.5/.6): -4.55 pp

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E6) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.